

PCSCHEMATIC Automation: Focus Only On Designing Your Electrical Systems

ASHWIN GOPINATH

Every design out there started off on a piece of paper. The glasses you might be wearing, the watch you do not use to tell the time anymore, the lampshade under which you are probably reading this article, everything. Even with the advent of design software, that first step has still retained its position. But you still need a way to bring that design out into the real world.

Designers today have a whole range of software to choose from to fulfil their every need, but they pay exorbitant amounts to purchase the licences enabling them to take advantage of the latest in the field, going well beyond conventional design. We would like to introduce you to an offering that shows how to get started with the same business, but without all the heavy machinery. Your first lesson, we feel, should not be designing the electrical framework for Rio Maracena, but something as fundamental as just lighting a bulb from kilometres away.

In this edition of *Electronics For You Plus* we will be taking a look at the electrical design tool Automation, the electrical CAD offering from PCSCHEMATIC. It is a free, advanced electrical

Features

- Exclusive workflow for electrical documentation, all in a single file
- Automatic updates for references to same component
- Intelligent electrical symbols
- Automatic PLC functions and reference overview symbols for PLCs
- Intelligent handling of reference designations
- Automatic renaming, replacing and numbering of symbols
- Creating projects by module-and-model based drag and drop

same philosophy. With a bunch of get-started videos on *YouTube*, and an extensive collection of articles to help you navigate the nook and crannies of this software, the problem of starting would not really be an issue.

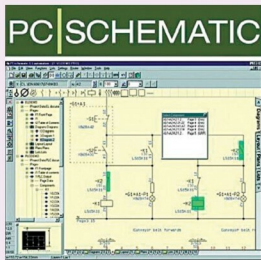


Fig. 1: PCSCHEMATIC Automation

CAD software for automation, installation, pneumatics as well as hydraulics. Not only is it easy on resources and complexity, it lets you maintain an overview of your designs while keeping the drawing part of the design simple.

Making the start easy

Automation is pretty easy to use, with an interface that follows the

The process of electrical system design

The process can be divided into seven steps, which would encompass most medium to large electrical systems. We start with a specification document, which is the specification and expectation list from the customer. A well-written document can mean that the same can be used through the entire cycle of design.

You can then create a functional specification document that goes into the more technical details of the project. This document is the basis for decisions made in this design.

Next, draw functional diagrams to indicate the next level of information; in this case, information and electrical power flow from one component to the other.

At this stage, employ schematic diagrams showing the interconnections between components, their job being to show a cursory hi-level look of the circuitry.